

Appliance Energy Forecasting

Time-Series Coding Case Study

Github Link : <https://github.com/Pampapathi25/appliance-energy-forecasting>

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Introduction

In this work, the energy consumption of the household appliances was predicted by increasingly elaborately developed time-series forecast techniques. The 10-minute data from UCI were Fourier transformed onto a 24-hour time frame, checked for accuracy of a linear trend at that time scale, and compared with measures of accuracy for 24-hour forecasting. Five benchmarks were accounted for, and subsequently an extensive search through the 147 possible combinations was conducted using SARIMAX AIC, a regression model based on the use of lag, rolling, calendar, sensor and the weather features, as well as the pre-trained Chronos-T5-small foundation model. The weekly seasonal naive model (RMSE 48.81) was the best indicator. The RMSE for XGBoost was 31.25 and the RMSE obtained by SARIMAX was the lowest one computed among the best comparable ones with 26.81 over a period of 24 hours. Chronos generated an RMSE of 56.76, MAE of 41.40 and a substantially negative bias of -40.02 with the added complexity which means that it wasn't worth the effort at this forecast origin. The results indicate that it is more important to include recent demand and repeat daily/weekly structure than to have the model scale alone. Given the current evidence, SARIMAX is recommended as it has the best accuracy observed, provides meaningful dynamics, and allows for the expression of forecast uncertainty.

1. Forecasting Problem

The response variable is energy consumption of the appliances per hour (in joules ('Appliances')). The primary job is to forecast the next day! With the feature-based model, the rest of the 14 days are held out for testing, and the first 24 hours constitute the next-day forecast. MAE, RMSE, MASE and Bias (where applicable) are used to perform accuracy assessment. It should be pointed out that large errors are significant for energy-demand planning, and MAE provides an interpretable mean absolute error, whereas the RMSE is emphasised.

2. Data Preparation and Exploratory Data Analysis

Total data collected from the low energy house is 19,735 ten-minute values from anew house comprising appliance demand along with indoor temperature/humidity value and outside weather data. Parses timestamp, validates data and combines measurements for the 10 minutes data interval into hourly data intervals. Hourly aggregation compares to the 24-step horizon of the forecast and is used to minimise the variability of measurements for very short range forecasts.

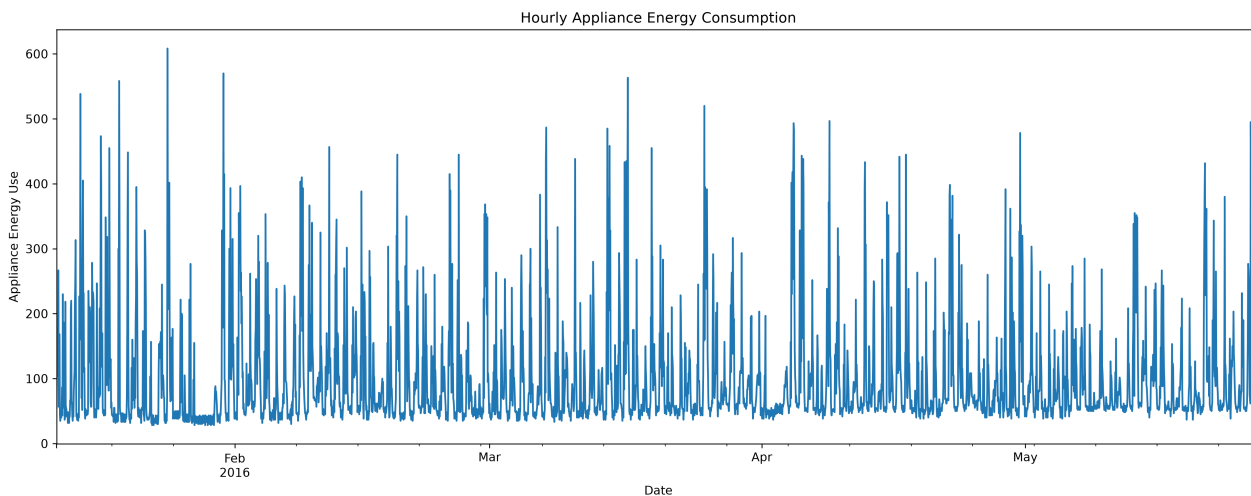


Figure 1. Hourly appliance energy consumption after preprocessing.

The full hourly series is extremely volatile and is marked by many transitory spikes. However, most observations are well below the peak, and there is a positive skew. This is important because a model can replicate the normal level very well but still will obtain a high RMSE for a high-demand event if it misses it.

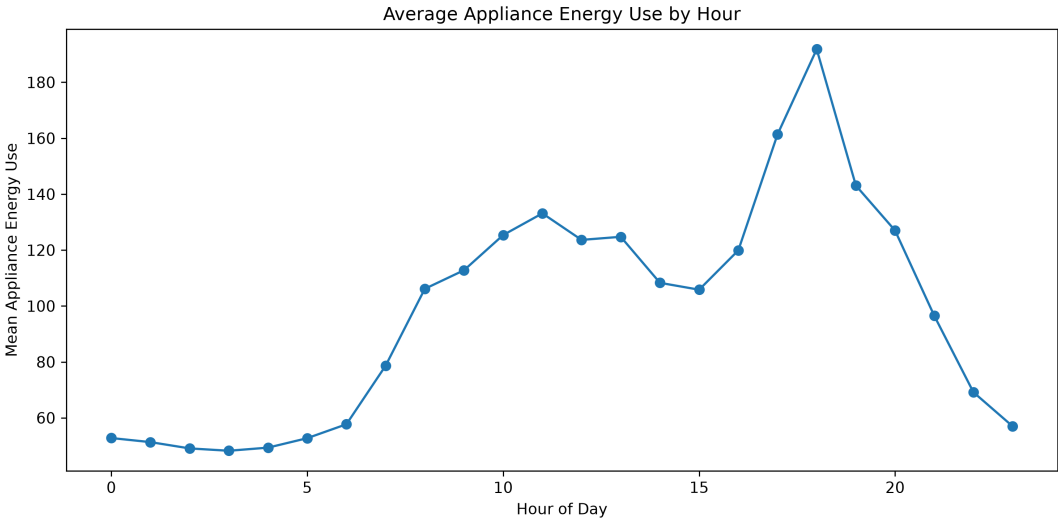


Figure 2. Average appliance energy use by hour of day.

The actual chart shows the daily trading cycles quite well in the form of an hourly profile chart. Consumption is at its lowest levels during the night hours and gradually increases through the morning and the largest average consumption occurs around 18:00, after which it drops. This adds a 24-hour time of the day (seasonal) dimension to daily periods that can be encouraged by cyclic features defined hour by hour (as in XGBoost).

3. Time-Series Components, Stationarity and Benchmarks

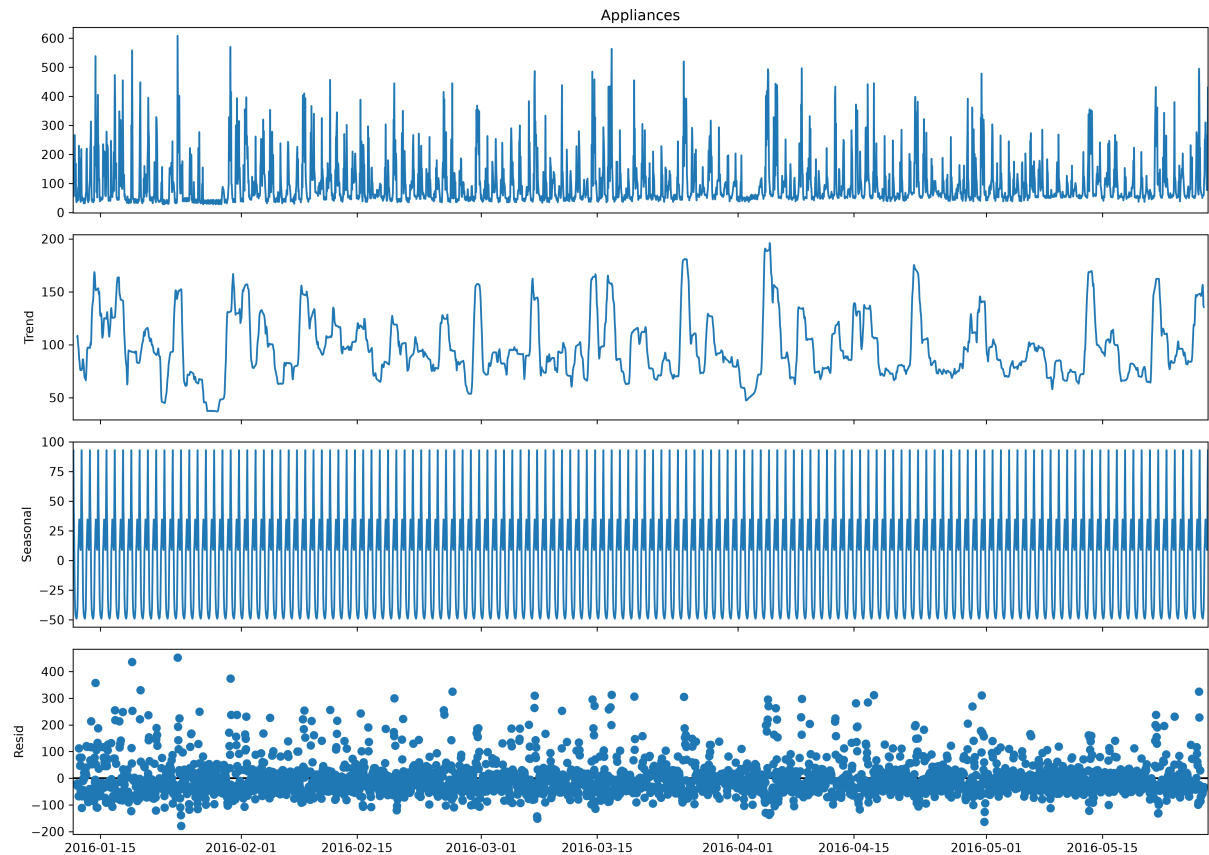


Figure 3. Additive decomposition using a 24-hour seasonal period.

This decomposition is splitting a repetitive seasonal component with almost identical daily signals from a residual component with irregular waves and significant positive spikes. The project also implements ACF/PACF analysis which looks at the serial dependence problem. ADF and KPSS tests were applied for checking the stationarity. The executed results yielded the statistic of approximately -8.95, p approximately 8.8e-15 by ADF and approximately 0.038, p=0.10 by KPSS. Together, these findings confirm that both these levels and both the catchment data are stationary and can be trusted as a starting point for d=0. Finally, there was a search for the d=0 AIC.

Benchmark forecasts

Model	MAE	RMSE	MASE	Bias
Mean	39.38	50.01	0.737	-8.23
Naive	242.64	247.60	4.542	242.64
Daily seasonal naive	80.28	118.04	1.503	58.89
Weekly seasonal naive	30.00	48.81	0.562	-6.53
Drift	243.88	248.76	4.565	243.88

The weekly seasonal naive model performs best compared to the requested benchmarks: naive, seasonal and drift models. It gives significantly better results than the daily seasonal naive (118.04) and simple naive/drift forecasts with its RMSE of 48.81. This indicates that the use of the appliances is very appealing to the behaviour from the previous and present hour of the same day, which is not true of the preceding days same hour and the preceding day same hour. Although the mean forecast outperforms weekly seasonal naive in terms of mean absolute error (MAE), both forecasts have similar root mean square errors (RMSE), and thus rankings based on the RMSE can vary.

4. SARIMAX Modelling, Selection and Diagnostics

SARIMAX was chosen due to high periodicity and high auto-correlation of the EDA. For the assignment, there are the three numbers p, d, and q and all possible combinations of p=0..6, d=0..2 and q=0..6, in all 147 candidate non-seasonal orders. These were fitted by Akaike information criterion and ranked by them. There are 147 rows in the committed search and two combinations of which no finite AIC was obtained. The lowest AIC was 32151.39 for order (2,0,6), followed closely by (4,0,6). AIC gets better with goodness of fit but worse with parameters — i.e. the chosen order is the best fitting yet simplest among searched orders in terms of likelihood/complexity.

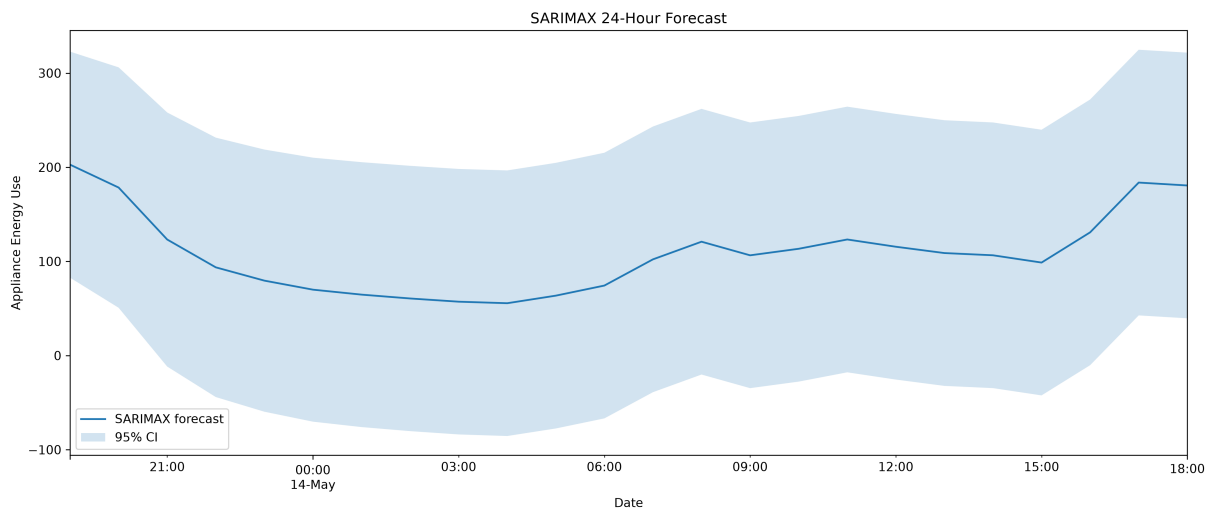


Figure 4. SARIMAX 24-hour forecast with 95% confidence interval.

SARIMAX model point forecast is able to mimic the expected overnight trend followed by day time rise. It has a large uncertainty and low values below 0 during the several hours. The interval clearly reveals a limitation of the gaussian linear formulation and not a possible physical result when looking at a negative energy use. A target that is changed or a non-negative probabilistic target will enhance realism.

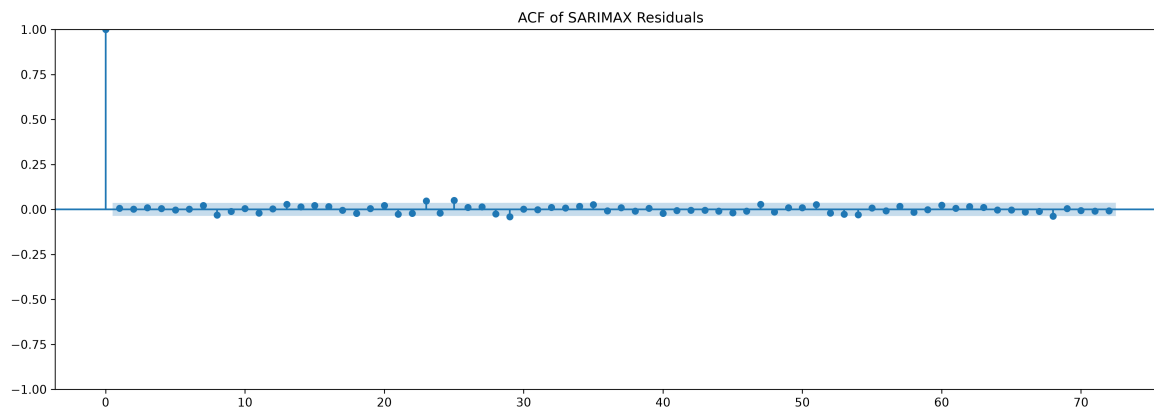


Figure 5. ACF of SARIMAX residuals.

There are small residual correlations. These residual independence p-values (Ljung-Box) are 0.335 at lag 24 or 3.5 years, and 0.319 at lag 48 or 7 years (not rejected at the 5% level for a seasonal horizon of 24 or 48 lags, or 3.5 or 7 years). Before matching the saved SARIMAX forecast with the actual data, PADJAlign and transform the data with the same 24 values with XGBoost, it is about MAE 18.93, RMSE 26.81 and Bias +3.26. This is the minimum RMSE compared to those currently being compared but these SARIMAX metrics should be saved as a part of the project, if reproducibility is a goal.

5. Feature Engineering and XGBoost

The feature-based model was designed to preserve non-linearities which a linear autoregressive model could not. Recent appliance vagars, day-of-week, day-of-month, hour-of-day, mean, standard deviation, rolling, indoor temperature and indoor humidity sensors, and outside weather sensors are engineered predictors. The final 14 days will be used as a test period and do NOT include random shuffling of future and past observations.

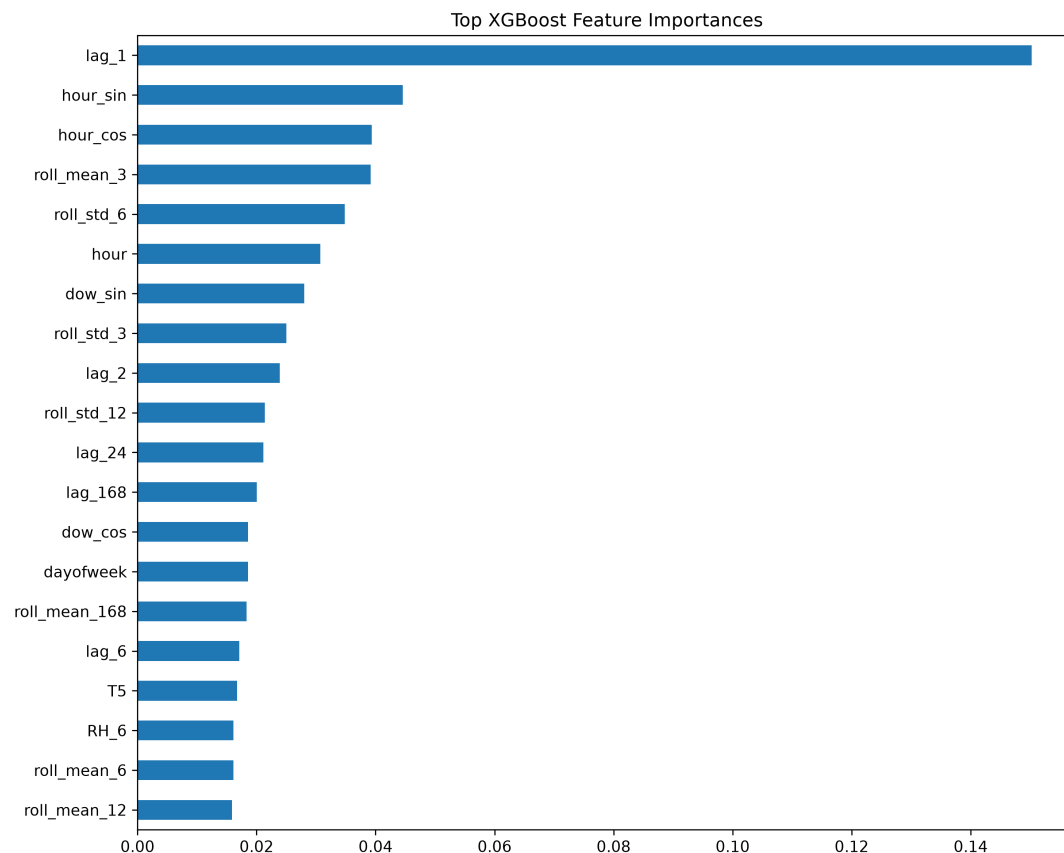


Figure 6. Top XGBoost feature importances.

By far the most important determinant, however, is lag_1, which demonstrates that there significant components of information in recent appliance demand. In addition, cyclic hour, short rolling windows, lag_24 and lag_168 are of note. This is in the context of EDA and benchmark characteristics: short memory dependence and daily/weekly recurrence in the series. The lag variables and calendar variables are the more significant variables in the top ranking, with the lag variables having the highest top ranking magnitude. The top ranking variables are sensor variables (T5 and RH_6) but their ranking magnitudes are less than the main lag variables and calendar variables.

XGBoost results

Model	MAE	RMSE	MASE	Bias
XGBoost	23.98	31.25	0.465	15.11

In comparison to the weekly seasonal benchmark RMSE (Root Mean Squared Error) with 48.81 drops down to 31.25, which represents a reduction of approximately 36%. Doing the same math for MASE, it has a value of 0.465 which is also below 1, or improvement over the output from the scaling baseline. But Bias is +15.11 with systematic overprediction for this 24-hour period. Feature importance is to be understood as usage in terms of predictiveness and not causality.

Covariate availability and leakage

Historical lags and historical features in the calendar are left in forecast origin. In comparison, in a real operating forecast, realised future values of the indoor environment (weather, temperature or humidity) would not be known in the test set. Those realised future observations makes the prediction conditional and can yield optimistic performance. Ideally, a deployment evaluation should only include variables known at the deployment site, or forecasts into the future should replace the actual weather/sensor data.

6. Chronos Foundation Model

The foundation-model stage requires the use of a pre-trained zero-shot time-series forecaster: Amazon Chronos-T5-small. Model takes the latest 168 (7-days) hourly observations and forecasts 24 hours ahead. A point forecast and a 90% prediction interval are calculated as the median of 100 of the forecast samples and the 5th and 95th quantile of the samples. This Chronos configuration instead uses the history of the target directly, instead of relying on the future sensor/weather covariates as in the XGBoost configuration.

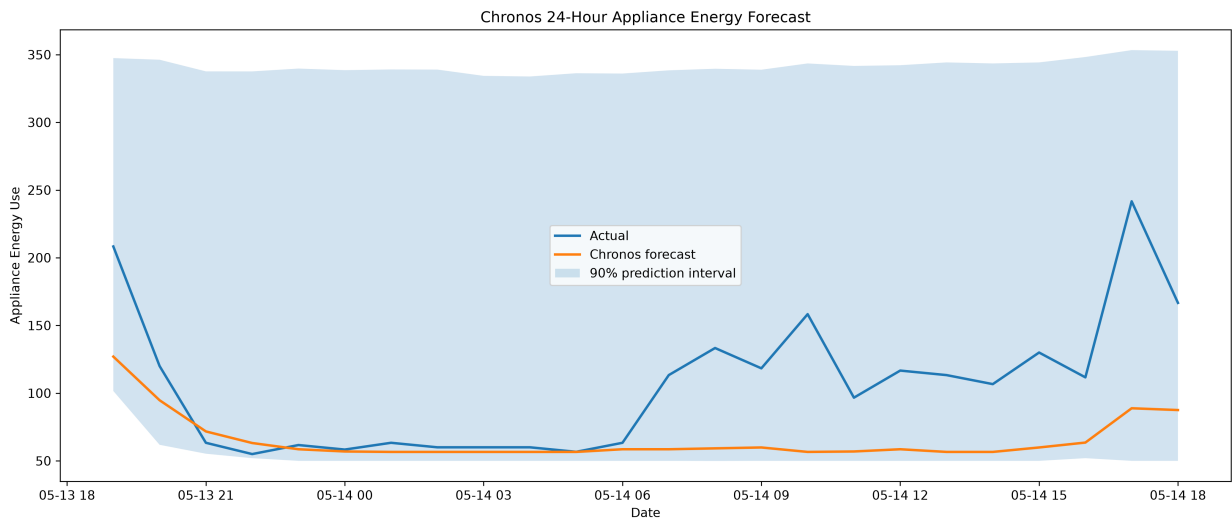


Figure 7. Chronos-T5-small 24-hour forecast with 90% prediction interval.

Model	MAE	RMSE	MASE	Bias	Context
Chronos-T5-small	41.40	56.76	0.775	-40.02	168 h

On this forecast origin chronos does not outperform the weekly seasonal benchmark, XGBoost or SARIMAX. The skill measures RMSE and MAE are 56.76 and 41.40, respectively, and the bias of -40.02 is large and significant in terms of systematic under-prediction. This can be explained by the mechanism displayed in the figure: Following a sharp downward trend, the Chronos median shows little ratio and is unable to replicate several ratio increases in the

series observed over time. The 90% interval is very broad and is sometimes larger than 330 and sometimes smaller than 50, so that it covers with a loss of resolution.

This finding is of significance as the result shows the foundation models are not necessarily better than specialised models for statistics or supervised learning in every dataset. The convenient zero-shot probabilistic forecasting by Chronos fails to capture the pre-trained representation, which is not ideally suited to the household-specific pattern seen at this origin. The accuracy is less good and the computational/dependency cost is greater here, and therefore the extra complexity is not worth it.

7. Overall Evaluation and Critical Discussion

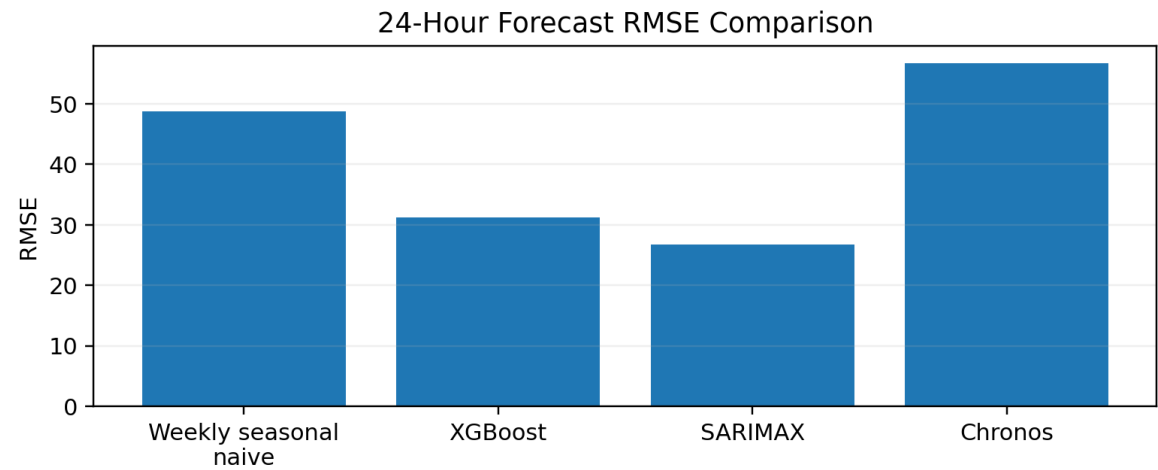


Figure 8. RMSE comparison for the strongest benchmark and three advanced models.

Model	MAE	RMSE	MASE	Bias	Rank by RMSE
SARIMAX	18.93	26.81	-	3.26	1
XGBoost	23.98	31.25	0.465	15.11	2
Weekly seasonal naive	30.00	48.81	0.562	-6.53	3
Chronos-T5-small	41.40	56.76	0.775	-40.02	4

Answers to the six required analysis questions

1. Which benchmark is strongest?

The strongest one of the above is the weekly one-seasonal naive because it includes only naive and weekly seasonal. This reflects a high level of weekly recurrence, and shows the value of establishing a seasonal baseline, rather than continuing the most recent observation.

2. Does SARIMAX improve on the strongest seasonal benchmark?

Yes. However, its RMSE is about 26.81 which is significantly lower than 48.81. There is good capture of daily seasonal dynamics, and an effective short-term autocorrelation is also obtained, with Ljung-Box tests at lags 24 and 48 indicating lack of significant autocorrelation. However, broad intervals and negative minimum values suggest that the uncertainty modelling is not doing a very good job.

3. Does the feature model improve with engineered variables?

Yes. XGBoost is doing a bit better (RMSE 31.25), with lag_1 at the top, followed by the cyclic hour features and the rolling features, and the final two confirming information on the daily and weekly scale. In addition to the fitted model, the sensor variables are also helpful but not as important as the demand-history and calendar features.

4. Does the foundation model outperform the simpler models?

The RMSE for No. Chronos is 56.76, which is poor compared to the weekly seasonal naive models, XGBoost and SARIMAX models. This improvement is negative, therefore, and the added model complexity is not justified for this experiment by the accuracy gained.

5. Which variables are genuinely known at forecast origin?

The time Delay-function of Historian-appliances and Calendar features are known. Realised future indoor sensor values are not. Forecasts for the future are not available as the test value but are forecasts. Realised future covariates is a conditional instead of true ex-ante forecast.

6. Which model should be recommended?

Based on current evidence, the recommendation is to use SARIMAX, since this model possesses minimum observed error, uncertainty and interpretable temporal dynamics. If there is a reliable covariate, XGBoost is a good option. Chronos is useful as a basis for making comparisons of an implemented model but not competitive enough to deploy here.

8. Limitations, Future Improvements and Conclusion

Limitations and future improvements

The experiment's one household and one principle 24 hour forecast origin is restrictive of generalisability. A better evaluation would be done with the help of rolling-origin back testing and would look at the error distribution over a wide variety of days (e.g. 100 days or more). XGBoost hyperparameters should be tuned by using time-series cross-validation; the ablation of feature-groups should be quantified by incorporating lags, rolling statistics, calendar variables, sensors and weather. The target of the SARIMAX can be converted to ensure it lies above the confidence limits and empirical interval coverage should be evaluated. In addition to testing on the newer/smaller versions of Chronos with the same forecast origin, alternate context length should also be taken into account. More importantly, the future covariates have to be processed in a leakage free process.

Reproducibility

The repository can store scripts, reusable source code, data, forecasts, metrics or figures. Preparation, EDA, benchmarks, SARIMAX search/diagnostics, feature engineering, XGBoost and Chronos are documented. The final submission must have the requirements.txt for the environment that you tested, all outputs must be committed to the repository, the pipeline must start from the project root directory and every numerical value reported here must be reproducible. The SARIMAX search and the diagnostic CSVs that were saved are helpful for providing a traceable history as to the parameters that were chosen.

Conclusion

The analysis shows that the structure in the energy use of such appliances is significant both across the day and across the week, and there is a noticeable short-term autocorrelation. Both SARIMAX and XGBoost give significant gains compared to simple weekly seasonal naive, which is the most informative simple benchmark. And as XGBoost does, the last few lags (and time of day) are the most important variables to be predictive. SARIMAX offers the best _accharity_ observed in 24 hours, has residual autocorrelation diagnostics which are also well behaved, but has uncertainty interval that requires physical caution. Chronos-T5-small essentially meets the intended goal for model comparison for the purposes of the foundation, but fails to accurately model the need and to outperform the simpler alternatives. The use of a model alone is not enough to ensure improved forecasts. From this experiment, SARIMAX is the best model for practical smart-home forecasting with the requirement of validation in every rolling forecasting origin.

References

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